Stochastic programming
Multistage problems

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Motivation: financial planning

(Birge and Louveaux, Section 1.2)

- We invest an initial wealth b, and would like to reach a goal G after Y years.
- The portfolio is rebalanced every v years: there are $H = Y/v$ investment periods, $\mathcal{T} = \{1, \ldots, H\}$, and an investment decision at the beginning of each period.
- N: set of available assets (stocks, bonds, ...).
- If the final wealth exceeds the goal G, the surplus earns interest at rate q.
- If it falls short of G, the shortfall must be borrowed at rate r (typically, $r > q$).

Variables and data

- x_{it}, $i \in N$, $t \in \mathcal{T}$: amount invested in asset i at the beginning of period t;
- ω_{it}: return of asset i during period t, i.e., the value at the end of the period of one dollar invested at its beginning;
- y: surplus over the goal at the end of the horizon;
- w: shortfall at the end of the horizon.

The decisions $x_{i2}, \ldots, x_{iH}$ rebalance the portfolio: they are corrective actions toward the goal G.

Deterministic formulation

$$
\begin{aligned}
\max\ & qy - rw \\
\text{s.t.}\ & \sum_{i \in N} x_{i1} = b, \\
& \sum_{i \in N} \omega_{i,t-1} x_{i,t-1} = \sum_{i \in N} x_{it}, \quad t = 2, \ldots, H, \\
& \sum_{i \in N} \omega_{iH} x_{iH} - y + w = G, \\
& x_{it} \geq 0, \quad i \in N,\ t \in \mathcal{T}, \qquad y, w \geq 0.
\end{aligned}
$$

The wealth at the end of period $t-1$ is entirely reinvested at the beginning of period t.

Random returns

- The returns are not known in advance: they are random variables.
- Assume that, in each period t, the vector of returns $(\omega_{it})_{i \in N}$ takes one of R possible values.
- A scenario is a sequence of realizations, one per period: there are R^H scenarios.

Example: $R = 4$ and $H = 2$, hence 16 scenarios:

$t = 1$	$t = 2$
1	1
1	2
1	3
1	4
2	1
⋮	⋮
4	4

Stochastic program formulation

- S: set of scenarios; p_s: probability of scenario $s \in S$;
- ω_{its}: return of asset i during period t in scenario s;
- x_{its}: amount invested in asset i at the beginning of period t in scenario s;
- y_s, w_s: surplus and shortfall at the end of the horizon in scenario s.

Each variable is copied once per scenario, and we maximize the expected value of $qy - rw$.

Stochastic version

$$\max \sum_{s\in S} p_s(qy_s - rw_s)$$

$$\text{s.t.} \sum_{i\in N} x_{i1s} = b, \quad \forall s \in S,$$

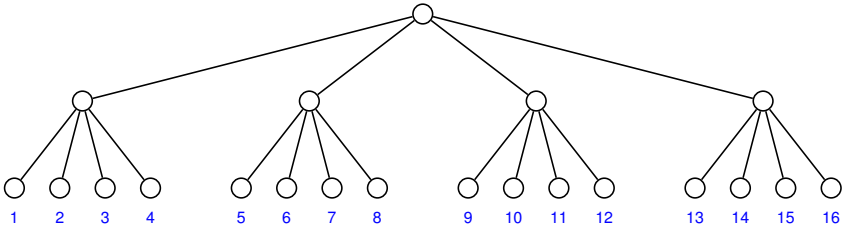
$$\sum_{i\in N} \omega_{i,t-1,s}\, x_{i,t-1,s} = \sum_{i\in N} x_{its}, \quad t = 2,\dots,H, \ \forall s \in S,$$

$$\sum_{i\in N} \omega_{iHs}\, x_{iHs} - y_s + w_s = G, \quad \forall s \in S,$$

$$x_{its} \geq 0, \quad i \in N, \ t \in \mathcal{T}, \ s \in S, \qquad y_s, w_s \geq 0, \quad s \in S.$$

Scenario tree

Conceptually, the sequence of events can be arranged in a tree: the root is the first decision, each branching is the realization of the returns of a period, and each leaf is a scenario. Example ($R = 4$, $H = 2$):



The scenarios can have different probabilities, and the returns of successive periods can be dependent: the branching probabilities can differ from one node to another.

Non-anticipativity

- Converting the deterministic model into the stochastic version is easy...
- but the resulting model is wrong: x_{its} may depend on the whole scenario s, including the returns of periods $t, \ldots, H$, unknown when x_{its} is decided. The model sees the future.
- We have to enforce non-anticipativity: a decision may only depend on the information available when it is made.

Let S_s^t be the set of scenarios identical to s up to period $t-1$, i.e., indistinguishable from s when the decisions of period t are made ($S_s^1 = S$). We impose

$$x_{its} = x_{its'}, \quad \forall i \in N,\ t \in \mathcal{T},\ s \in S,\ s' \in S_s^t.$$

The final y_s and w_s are decided once everything is known: they need no such constraint.

Non-anticipativity (cont'd)

Equivalently, as every $p_s > 0$,

$$\Big(\sum_{s' \in S_s^t} p_{s'}\Big) x_{its} = \sum_{s' \in S_s^t} p_{s'} x_{its'}, \quad \forall i, t, s,$$

that is,

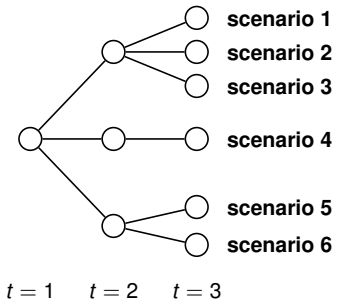
$$x_{its} = \frac{\sum_{s' \in S_s^t} p_{s'} x_{its'}}{\sum_{s' \in S_s^t} p_{s'}} = \mathbb{E}\left[x_{its'} \mid s' \in S_s^t\right], \quad \forall i, t, s.$$

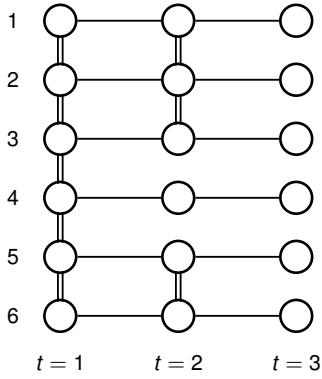
Each decision equals its conditional expectation given the information available when it is made: if all the decisions of a group of indistinguishable scenarios equal the group average, they are all equal. This form underlies progressive hedging (Lagrangian methods slides).

There are many other formulations, for instance equalities between consecutive scenarios of each group (next slide).

Back to the scenario tree

In the scenario formulation, each decision is copied once per scenario, and non-anticipativity ties the copies together.





Right: one row per scenario, one column per period; a double bar is an equality $x_{its} = x_{its'}$ between two consecutive scenarios.

Nodal approach

- Alternatively, define one vector of decisions per node of the tree: the decision made given the realizations observed up to this node.
- Non-anticipativity is then built in: all the scenarios going through a node share its decision.
- $\mathcal{L}$: set of decision nodes, with root 1; the leaves are the scenarios $s \in S$.
- $a(\ell)$: parent of node ℓ (or of leaf s), i.e., its ancestor at the previous stage; $\omega_{i\ell}$: return of asset i on the branch from $a(\ell)$ to ℓ.
- Fewer variables, $|N|\,|\mathcal{L}|$ instead of $|N|\,H\,|S|$, and no non-anticipativity constraint.

Nodal formulation

$$
\begin{aligned}
\max \quad & \sum_{s \in S} p_s (q y_s - r w_s) \\
\text{s.t.} \quad & \sum_{i \in N} x_{i1} = b, \\
& \sum_{i \in N} \omega_{i\ell}\, x_{i,a(\ell)} = \sum_{i \in N} x_{i\ell}, \quad \forall \ell \in \mathcal{L} \setminus \{1\}, \\
& \sum_{i \in N} \omega_{is}\, x_{i,a(s)} - y_s + w_s = G, \quad \forall s \in S, \\
& x_{i\ell} \geq 0, \quad i \in N,\ \ell \in \mathcal{L}, \qquad y_s, w_s \geq 0, \quad s \in S.
\end{aligned}
$$

Multi-period production planning

- A firm produces several goods over several periods.
- Production uses resources (for instance, labor and machine hours); the amount of each resource needed per unit of each product is known.
- The demand of each period is random, and must be met: missing products are bought from an external source, and unsold ones are stored. Both have a cost.
- Additional resources (overtime, machine hours) can be bought, within limits.
- Changes in the workforce from one period to the next have a cost (hiring, firing).

Decision problem

- We decide now, for every period:
  - the quantity of each product to manufacture;
  - the additional resources to use;
  - the hiring and firing of staff.
- At each period, the random demand is observed.
- Then, we decide how much to store, and how much to buy from the external source.

The production plan is decided once, but the storage and purchase decisions of period t may only use the demands of periods $1, \ldots, t$: the problem is multistage.

Sets and variables

1. Sets:
   - $\mathcal{T} = \{1, \ldots, H\}$: periods;
   - N: products;
   - M: resources; $L \in M$ is the labor.
2. Variables:
   - x_{jt}: production of product $j \in N$ in period $t \in \mathcal{T}$;
   - u_{it}: additional amount of resource $i \in M$ bought for period t;
   - z_t^+, z_t^-: increase and decrease of the workforce from period $t-1$ to period t;
   - y_{jt}^+: surplus of product j at the end of period t, stored; y_{jt}^-: deficit, bought from the external source.

Parameters

- α_{jt}, β_{it}, γ_t^+, γ_t^-, δ_{jt}^+, δ_{jt}^-: unit costs of the variables above;
- ω_{jt}: random demand of product $j \in N$ in period $t \in \mathcal{T}$;
- a_{ij}: amount of resource $i \in M$ required to produce one unit of product $j \in N$;
- b_{it}: amount of resource $i \in M$ available in period $t \in \mathcal{T}$;
- U_{it}: upper bound on u_{it};
- y_{j0}^+: initial inventory of product j.

Model

$$\min \sum_{j\in N}\sum_{t\in\mathcal{T}} \alpha_{jt}x_{jt} + \sum_{i\in M}\sum_{t\in\mathcal{T}} \beta_{it}u_{it} + \sum_{t=2}^{H}\left(\gamma_t^+ z_t^+ + \gamma_t^- z_t^-\right) + \mathbb{E}_\omega\left[\min \sum_{j\in N}\sum_{t\in\mathcal{T}}\left(\delta_{jt}^+ y_{jt}^+ + \delta_{jt}^- y_{jt}^-\right)\right]$$

$$\text{s.t.} \sum_{j\in N} a_{ij}x_{jt} \le b_{it} + u_{it}, \quad 0 \le u_{it} \le U_{it}, \quad \forall i \in M,\ t \in \mathcal{T},$$

$$z_t^+ - z_t^- = \sum_{j\in N} a_{Lj}(x_{jt} - x_{j,t-1}), \quad t = 2,\dots,H, \qquad x, z^\pm \ge 0,$$

$$x_{jt} + y_{j,t-1}^+ - y_{jt}^+ + y_{jt}^- = \omega_{jt}, \quad \forall j \in N,\ t \in \mathcal{T}, \qquad y^\pm \ge 0 \quad \text{(demand satisfaction)}.$$

Inner min: $y_{jt}^\pm$ depends only on the demands up to period t.

Extended form of the recourse model

$$\min \sum_{j\in N}\sum_{t\in\mathcal{T}} \alpha_{jt}x_{jt} + \sum_{i\in M}\sum_{t\in\mathcal{T}} \beta_{it}u_{it} + \sum_{t=2}^{H}\left(\gamma_t^+ z_t^+ + \gamma_t^- z_t^-\right) + \sum_{s\in S} p_s \sum_{j\in N}\sum_{t\in\mathcal{T}} \left(\delta_{jt}^+ y_{jts}^+ + \delta_{jt}^- y_{jts}^-\right)$$

$$\text{s.t.} \sum_{j\in N} a_{ij}x_{jt} \le b_{it} + u_{it}, \quad 0 \le u_{it} \le U_{it}, \quad \forall i \in M,\ t \in \mathcal{T},$$

$$z_t^+ - z_t^- = \sum_{j\in N} a_{Lj}(x_{jt} - x_{j,t-1}), \quad t = 2,\dots,H, \qquad x, z^{\pm} \ge 0,$$

$$x_{jt} + y_{j,t-1,s}^+ - y_{jts}^+ + y_{jts}^- = \omega_{jts}, \quad \forall j \in N,\ t \in \mathcal{T},\ s \in S, \qquad y^{\pm} \ge 0,$$

$$y_{jts}^{\pm} = y_{jts'}^{\pm} \quad \text{if } s \text{ and } s' \text{ have the same demands up to period } t.$$

Potential solution techniques

Adapted from Jalal Kazempour,
`https://www.jalalkazempour.com/teaching`

- Direct solution: solve the extensive form with a linear programming solver; its size grows exponentially with the number of stages.
- Linear decision rules: restrict each decision to an affine function of the uncertain parameters observed so far. This approximation (an upper bound, for a minimization) needs no decomposition: the size of the problem grows only polynomially with the number of stages.
- Nested decomposition (nested Benders): fixes the decisions of the earlier stages (complicating variables), and decomposes the problem by node of the tree.
- Lagrangian relaxation: relaxes the non-anticipativity constraints (complicating constraints), and decomposes the problem by scenario.

Nested decomposition

The nested L-shaped method (Birge, 1985) applies the L-shaped method recursively on the scenario tree, to solve

$$\min_{x_1 \geq 0} \; c_1^T x_1 + \mathbb{E}\Big[\min_{x_2 \geq 0} c_2^T x_2 + \mathbb{E}\big[\cdots + \mathbb{E}[\min_{x_H \geq 0} c_H^T x_H]\big]\Big]$$

subject to $W_1 x_1 = h_1$ and $W_t x_t = h_t(\xi_t) - T_{t-1}(\xi_t) x_{t-1}$, $t = 2, \ldots, H$. The decision x_t is made once $\xi_{[t]} = (\xi_2, \ldots, \xi_t)$ is known, before ξ_{t+1} (non-anticipativity).
Cost-to-go functions, with $\mathcal{Q}_{H+1} \equiv 0$:

$$V_t(x_{t-1}, \xi_{[t]}) = \min_{x_t \geq 0} \Big\{ c_t^T x_t + \mathcal{Q}_{t+1}(x_t, \xi_{[t]}) \;:\; W_t x_t = h_t(\xi_t) - T_{t-1}(\xi_t) x_{t-1} \Big\},$$
$$\mathcal{Q}_{t+1}(x_t, \xi_{[t]}) = \mathbb{E}\left[V_{t+1}(x_t, \xi_{[t+1]}) \mid \xi_{[t]} \right].$$

On a finite tree, each $\mathcal{Q}_{t+1}(\cdot, \xi_{[t]})$ is convex and piecewise linear. The SDDP slides give the details of the method.

Nested decomposition: cuts

Node k at stage t: master problem for its children, subproblem for its parent $a(k)$:

$$\min_{x_{t,k},\,\theta_{t,k}} \; c_{t,k}^T x_{t,k} + \theta_{t,k} \quad \text{s.t. } W_t x_{t,k} = h_{t,k} - T_{t-1,k}\,\hat{x}_{t-1,a(k)}, \quad (\pi_{t,k})$$

$$\text{feasibility and optimality cuts,} \quad x_{t,k} \geq 0,$$

where $\hat{x}_{t-1,a(k)}$ is the parent's trial decision, and $\theta_{t,k}$ underestimates the expected cost-to-go at node k ($\theta_{H,k} \equiv 0$). With $v_{t,k}$ the optimal value, the optimality cut of a node j at stage $t-1$ averages the supporting hyperplanes of its children $k \in \mathcal{D}_t(j)$:

$$\theta_{t-1,j} \geq \sum_{k \in \mathcal{D}_t(j)} p(k, t \mid j, t-1) \left(v_{t,k} - \pi_{t,k}^T T_{t-1,k} \left(x_{t-1,j} - \hat{x}_{t-1,j} \right) \right).$$

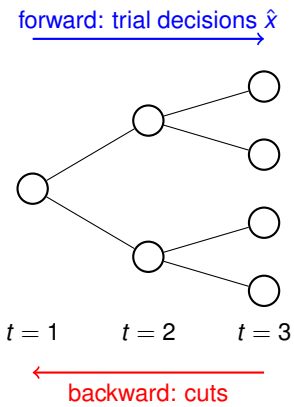
$v_{t,k}$ includes the children's own cuts: information propagates up the tree. These values underestimate the true ones, so the cut is valid.

Nested decomposition: forward and backward passes

Forward pass: solve the first-stage problem, send $\hat{x}_1$ to its children, solve them, and so on down to stage H. If every problem is feasible, these trial decisions form a policy, whose expected cost is an upper bound.

Backward pass: from stage H up to stage 2, each node receives a cut from its children (a feasibility cut if one is infeasible), is solved again, and sends a cut to its parent. The first-stage value is a lower bound.

Stop when the bounds meet. On a finite tree, with bounded x_t, only finitely many distinct cuts exist: finite convergence (Birge and Louveaux, Section 6.1, Theorem 1).



Beyond nested decomposition

The scenario tree grows exponentially with the number of stages, and so does the number of problems solved at each iteration of nested decomposition.

- Stochastic dual dynamic programming (Pereira and Pinto, 1991; SDDP slides): under stagewise independence, all the nodes of a stage share the same expected cost-to-go function, hence the same cuts, and the forward pass samples scenarios instead of enumerating the tree. Extensions handle Markovian processes, on a lattice.
- Lagrangian methods (progressive hedging; Lagrangian methods slides): relax the non-anticipativity constraints, and decompose the problem by scenario, each subproblem being a deterministic multistage problem. They do not require a Markovian process, but there is one subproblem per scenario, and many iterations can be needed to restore non-anticipativity.